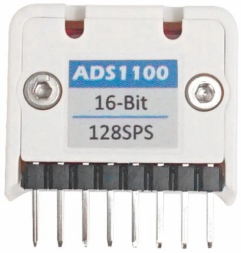
Hat ADC

SKU:U069

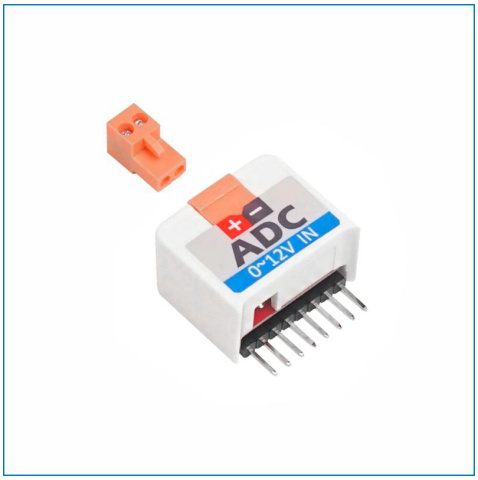














Description

Hat ADC is a high-precision analog-to-digital conversion module designed specifically for the StickC series. It integrates the ADS1100 16-bit Δ-Σ ADC chip, featuring fully differential input, self-calibration, and programmable gain. The chip itself supports differential inputs from -5V ~ +5V. Through optimized peripheral circuit design, the module extends this capability to a DC voltage measurement range of 0 ~ 12V. Using a standard I2C communication interface, the module enables accurate and stable voltage acquisition, making it suitable for industrial sensor signal processing, battery voltage monitoring, and various analog signal acquisition applications.

Features

- Input Voltage: 0 ~ 12V
- ADS1100:
  - 16-bit no-missing-codes resolution
  - Continuous self-calibration
  - Single-cycle conversion
  - Built-in programmable gain amplifier (Gain = 1, 2, 4, 8)
  - Low noise: 4μVp-p
  - Programmable data rate: 8SPS to 128SPS
  - Internal system clock
  - I2C communication interface (0x48)
- Development Platform
  - UiFlow1
  - UiFlow2
  - Arduino IDE

Includes

- 1 x Hat ADC
- 1 x HT3.96-2P Terminal
- 1 x Double-Sided Tape

Applications

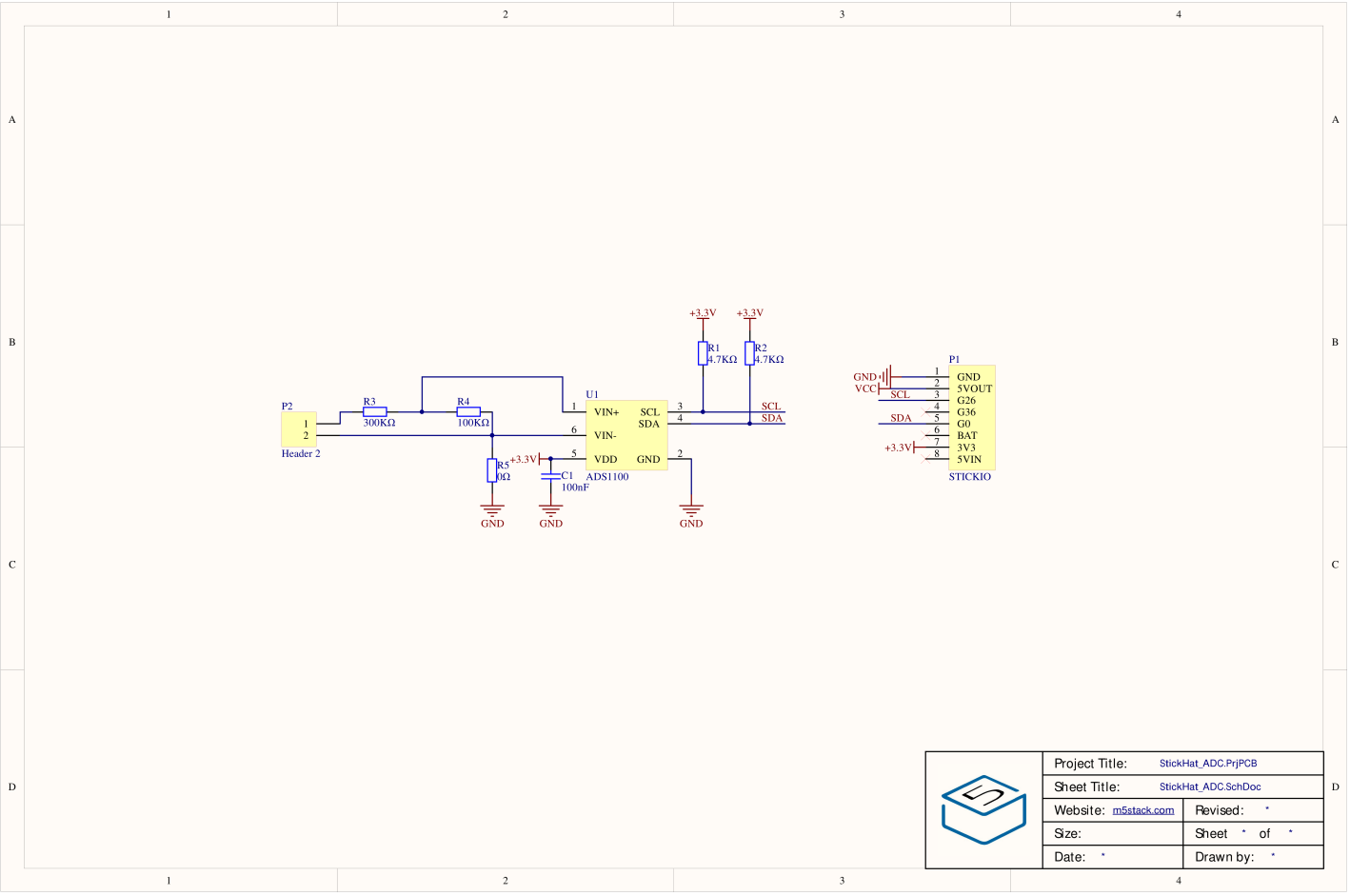
- Analog signal capture

Specifications

Specification	Parameter
Communication Interface	I2C Communication @0x48
Product Size	24.9 x 24.0 x 13.7mm
Product Weight	6.0g
Package Size	138.0 x 93.0 x 15.0mm
Gross Weight	14.0g

Schematics

- Hat ADC Schematics PDF

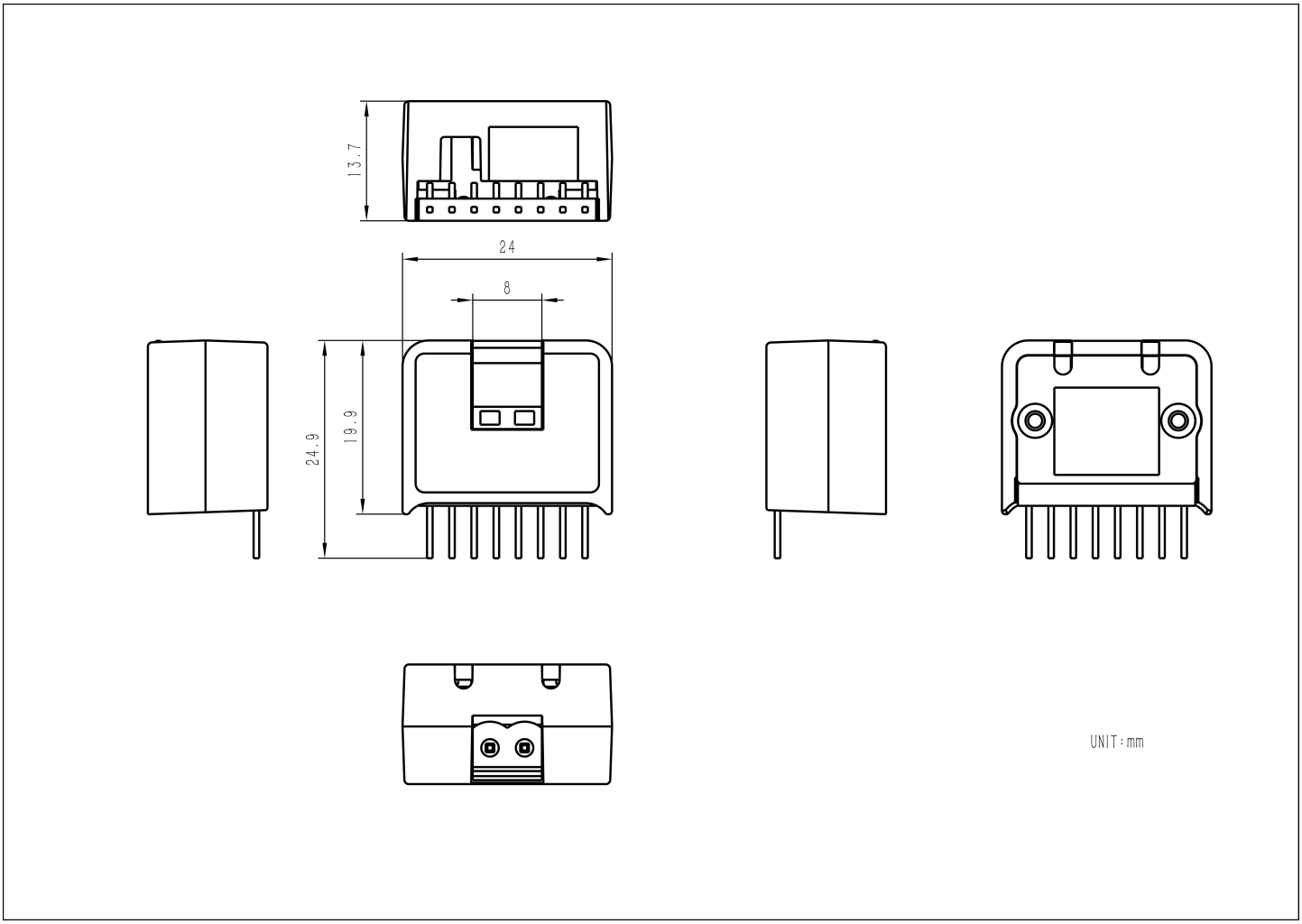


PinMap

M5StickC	G0	G26	5V	GND
HAT ADC	SDA	SCL	5V	GND

Model Size

Hat ADC Model Size PDF



Structure

- Hat ADC Structure Files

Datasheets

- ADS1100

Softwares

Arduino

- Hat ADC Example - with M5StickC
- Hat ADC Example - with M5StickC-Plus

UiFlow1

- Hat ADC UiFlow1 Docs

UiFlow2

- Hat ADC UiFlow2 Docs

Easyloader

Easyloader	Download	Note
Hat ADC Easyloader	download	/

Video

ADC-DAC-HAT.mp4